FROM THE PUBLICATION

Two imaging sub-endpoints, published. The quantitative ultrasound and gait-kinematics readouts summarized here are objective sub-endpoints reported in Bertone et al., Front. Vet. Sci. 2026 (DOI 10.3389/fvets.2026.1833933).

SCIENTIFIC REPORT · IMAGING & KINEMATICS · BERTONE ET AL. 2026

Two objective measures. One consistent signal.

Quantitative ultrasound tissue scoring and smartphone gait kinematics each showed greater recovery with eCTM than standard of care.

Background

OBJECTIVE

Clinical exam and owner report are inherently subjective. We paired two independent, objective modalities — quantitative ultrasound tissue scoring and smartphone-based gait kinematics — to ask whether eCTM produced measurable structural and functional recovery beyond what clinical scoring alone could capture.

Methods

QUANTITATIVE ULTRASOUND (TRS)

B-mode cine loops (Butterfly iQ3) at Weeks 0, 2, 4 and 12. Six texture features combined into a composite Tissue Remodeling Score (TRS), normalized to 50 at baseline. Thresholds: Responding (55+) · Equivocal (45–55) · Not Responding (<45). Segmentation was blinded to treatment.

GAIT KINEMATICS (SLEIP)

Smartphone video captured at Weeks 0, 2, 4 and 12 and analyzed on a dedicated motion-analysis platform (Sleip.com), yielding asymmetry indices for the impact and push-off phases of stride. Forelimb data were the most consistent and least variable; hindlimb and combined-limb data were more variable and underpowered for further between-group analysis.

Results

ULTRASOUND TRS · LIGAMENTS REACHING RESPONDING BY WEEK 12

36% : 0%

8 / 22 eCTM vs. 0 / 11 standard of care
$p = 0.013$ · Cohen's $d = 0.98$

GAIT KINEMATICS (SLEIP) · SIGNIFICANT IMPROVEMENTS FAVORING eCTM

METRIC	TIME	RESULT
Forelimb impact	Week 4	**$p < 0.05$**
Forelimb push-off	Week 2	**$p < 0.003$**
Forelimb push-off	Week 12	**$p < 0.01$**
Combined fore + hind, impact[1]	Week 12	**$p < 0.026$**

1. Combined fore + hind data were more variable than forelimb-only data and were flagged in the manuscript as underpowered for further between-group analysis.

Interpretation

TWO MODALITIES, ONE DIRECTION

Two independent, objective modalities pointed the same direction. By Week 12, more than a third of eCTM ligaments reached the Responding TRS threshold versus none in the standard-of-care group, and smartphone-based gait kinematics showed statistically significant forelimb improvement with eCTM at multiple time points for both impact and push-off. Forelimb data were the most consistent kinematic signal; combined fore + hind results moved the same direction but came from a smaller, more variable dataset and should be read as supportive, not standalone, evidence.

Evidence you can see, twice over.

Equine Performance Labs

Objective imaging and gait data reinforced the clinical findings — eCTM horses showed measurable structural and functional recovery that standard of care did not match.